SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE:CSA17**

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I **S.Rathika(192419042)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.
*Signature of the student: **S.Rathika***

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

1. Propositional Logic

Let:

R = It is raining.

W = The ground is wet.

Statement 1: $R \rightarrow W$

Statement 2: R

Goal: Prove W.

2. Convert to CNF

$R \rightarrow W \equiv \neg R \lor W$

CNF clauses:

C1: $\neg R \lor W$

C2: R

Negated goal: C3: $\neg W$

3. Resolution Steps

C1: $\neg R \lor W$
C2: R

C4: W

C4: W
C3: $\neg W$

□

4. Conclusion
Since the empty clause □ is derived, the negation of the goal leads to a contradiction. Therefore, the ground is wet.

Resolution tree:

```
¬R ∨ W      R
   \      /
     W
     |
    ¬W
     |
     □
```

Question 2 – Student Assignment Submission

1. Propositional Logic
Let:

S = Rahul submitted the assignment.

M = Rahul receives internal marks.

Statement 1: $S \rightarrow M$

Statement 2: S

Goal: Prove M.

2. CNF Conversion
$S \rightarrow M \equiv \neg S \lor M$

C1: $\neg S \lor M$

C2: S

Negated goal: C3: $\neg M$

3. Resolution
$\neg S \lor M$
S

M

M
$\neg M$

□

4. Conclusion
The empty clause □ is obtained. Therefore, Rahul receives internal marks.

Question 3 – Library Membership

1. Propositional Logic
Let:

L = Priya is a library member.

B = Priya can borrow books.

Statement 1: $L \rightarrow B$

Statement 2: L

Goal: Prove B.

2. CNF Conversion
$L \rightarrow B \equiv \neg L \vee B$

C1: $\neg L \vee B$

C2: L

Negated goal: C3: $\neg B$

3. Resolution
$\neg L \vee B$
L

B

B
$\neg B$

□

4. Final Conclusion
Since □ is derived, the assumption that Priya cannot borrow books is contradictory. Therefore, Priya can borrow books.

Question 4 – Placement Eligibility

1. Propositional Logic
Let:

A = Arun cleared the aptitude test.

E = Arun is eligible for placement.

Statement 1: $A \rightarrow E$

Statement 2: A

Goal: Prove E.

2. CNF Clauses
$A \rightarrow E \equiv \neg A \lor E$

C1: $\neg A \lor E$

C2: A

Negated goal: C3: $\neg E$

3. Resolution Steps
C1: $\neg A \lor E$
C2: A

C4: E

C4: E
C3: $\neg E$

□

4. Resolution Tree

```
¬A ∨ E       A
   \       /
    \     /
      E
      |
```

$\neg E$
|
□

5. Conclusion
The empty clause □ is derived. Therefore, Arun is eligible for placement.

Question 5 – Access Control System

1. Propositional Logic
Let:

P = The user entered the correct password.

A = The user is authenticated.

G = The user is granted access.

Statement 1: $P \rightarrow A$

Statement 2: $A \rightarrow G$

Statement 3: P

Goal: Prove G.

2. CNF Conversion
$P \rightarrow A \equiv \neg P \lor A$

$A \rightarrow G \equiv \neg A \lor G$

C1: $\neg P \lor A$

C2: $\neg A \lor G$

C3: P

Negated goal: C4: $\neg G$

3. Resolution Steps
Step 1: Derive Authentication

C1: $\neg P \lor A$
C3: P

C5: A

Step 2: Derive Access

C2: ¬A ∨ G
C5: A

C6: G

Step 3: Derive the Empty Clause

C6: G
C4: ¬G

□

4. Resolution Tree

```
¬P ∨ A        P
   \         /
    \       /
       A
       |
    ¬A ∨ G
       |
       G
       |
      ¬G
       |
       □
```

5. Final Conclusion
The resolution process derives the empty clause □. Therefore, the negation of the goal (¬G) is inconsistent with the knowledge base. Hence, the user is granted access.

Overall Resolution Principle
1. Represent the statements using propositional variables.

2. Convert implications into CNF using $P \rightarrow Q \equiv \neg P \lor Q$.

3. Add the negation of the goal to the clause set.

4. Resolve clauses to derive new clauses.

5. If the empty clause (□) is derived, the negated goal is contradictory.

6. Therefore, the original goal is proved.

Summary

Question	Goal	Empty Clause Derived?	Conclusion
1. Rain and Wet Ground	Ground is wet	Yes	Proved
2. Assignment Submission	Rahul receives internal marks	Yes	Proved
3. Library Membership	Priya can borrow books	Yes	Proved
4. Placement Eligibility	Arun is eligible	Yes	Proved
5. Access Control	User is granted access	Yes	Proved